

ASSIGNMENT-1

1. TensorFlow:

Distinct Features:

- TensorFlow is an open-source deep learning framework developed by Google.
- It offers a comprehensive ecosystem for machine learning and deep learning.
- TensorFlow provides both high-level APIs (like Keras) and low-level APIs for maximum flexibility.
- It supports distributed computing and can run on various platforms, including CPU and GPU.
- Functionality:
- TensorFlow is known for its computational graph-based approach, making it highly efficient for deep learning.
- It has extensive libraries for neural networks, including pre-built layers and optimizers.
- TensorFlow also supports deployment on mobile and embedded devices, making it versatile.

2. Keras:

Distinct Features:

- Keras is an open-source high-level neural networks API that can run on top of TensorFlow, Theano, or other deep learning frameworks.
- It is designed for easy and rapid prototyping of neural networks.
- Keras is user-friendly, with a simple and consistent API.
- Functionality:
- Keras is excellent for beginners and for quickly building and testing neural network models.
- It provides a high-level interface for defining and training neural networks with minimal code.
- Keras can seamlessly switch between different backend engines, such as TensorFlow and Theano.

3. Theano:

Distinct Features:

- Theano is an open-source numerical computation library that specializes in optimizing mathematical expressions.
- It is one of the early deep learning frameworks and has been influential in the field.
- Theano focuses on CPU and GPU computation optimization.
- Functionality:
- Theano is used as a backend for other deep learning frameworks like Keras.
- It is highly efficient for symbolic mathematics and can automatically optimize expressions for better performance.
- While Theano has been influential, it is less actively developed now in favor of TensorFlow and PyTorch.

4. PyTorch:

Distinct Features:

- PyTorch is an open-source deep learning framework developed by Facebook's AI Research lab.
- It's known for its dynamic computation graph, which makes it more flexible for certain applications.
- PyTorch has gained popularity for research and rapid experimentation.
- Functionality:
- PyTorch's dynamic computation graph allows for easy debugging and dynamic changes during runtime.
- It is highly popular in the research community due to its flexibility and support for dynamic networks.
- PyTorch also offers a wide range of libraries for natural language processing, computer vision, and reinforcement learning.